E-Commerce Sales & Customer Behaviour Analysis

1. Problem Statement

E-commerce transaction data may contain missing values, inconsistent categories, invalid dates, duplicate records and extreme values. These issues can affect the reliability of business analysis. This project uses Python to clean the dataset, perform exploratory data analysis (EDA), identify sales and profitability patterns, and provide practical business recommendations.

2. Dataset Overview

The project uses *ecommerce_dirty_dataset.csv*, which initially contained 100,500 rows and 16 columns. After the documented cleaning steps and duplicate removal, the dataset contained 99,518 rows and 16 columns.

Column Group	Columns	Purpose
Order details	Order_ID, Order_Date, Ship_Date	Track orders and shipping dates.
Customer and location	Customer_ID, Customer_Segment, Region, State	Analyse customer segments and geographic performance.
Product details	Product_Category, Sub_Category, Product_Name	Compare product and category performance.
Financial details	Sales, Quantity, Discount, Profit, Shipping_Cost	Analyse sales, discounts, profit and shipping costs.
Payment	Payment_Mode	Review payment methods.

3. Data Cleaning and Transformation

1. Data Type Conversion: Converted Order_Date and Ship_Date to datetime format and Sales and Profit to numeric format for accurate analysis.

2. Duplicate Removal: Used df.drop_duplicates() to remove exact duplicate records and prevent duplicate data from affecting results.

3. Handling Inconsistent Categories: Used str.strip() and str.title() to remove unwanted spaces and standardise product category names.

4. Handling Invalid Dates: Identified and removed records where Ship_Date was earlier than Order_Date, as these violate the expected shipping sequence.

5. Handling Negative Sales: Excluded records with negative Sales values based on the cleaning rules used in this project.

6. Handling Missing Customer Segment Values: Replaced missing Customer_Segment values with Unknown to retain records without assigning an unsupported category.

7. Handling Missing Profit Values: Left missing Profit values unchanged. Mean or median imputation was avoided because it could introduce unsupported financial estimates and potentially create inconsistent relationships between Sales and Profit.

8. Handling Missing Shipping Cost Values: Filled missing Shipping_Cost values with the median to retain records while limiting the influence of extreme values.

9. Handling Shipping Cost Outliers: Applied the Interquartile Range (IQR) method to calculate the upper limit and capped extreme Shipping_Cost values.

10. Handling Profit Outliers: Identified extreme Profit values using the IQR method, removed records where Profit equalled 500,000, and capped the remaining upper outliers.

4. Feature Engineering

Created date-based features from the order and shipping dates to support time-based and operational analysis.

- **Order_Year:** Extracted the year from Order_Date to compare annual sales performance.
- **Order_Month:** Extracted the month from Order_Date to analyse monthly sales trends.
- **Order_Quarter:** Extracted the quarter from Order_Date to compare quarterly business performance.
- **Shipping_Days:** Calculated the difference between Ship_Date and Order_Date in days to analyse shipping duration.

5. Exploratory Data Analysis (EDA) and Key Insights

1. Univariate Analysis: Reviewed the distributions of Sales, Profit, Discount and Shipping_Cost using descriptive statistics, histograms and KDE curves to understand spread, skewness and potential outliers.

2. Bivariate Analysis: Used scatter plots and correlation analysis to examine numerical relationships. Sales and Profit had a strong positive correlation of 0.89, while Shipping_Cost and Profit had almost no linear correlation.

3. Categorical Analysis: Compared sales and profit by region, product category and customer segment. The North region recorded the highest sales, and Electronics generated the highest sales and total profit among product categories.

4. Discount and Profitability Analysis: Average profit declined from approximately 11,564 at 0% discount to 7,184 at 50% discount, suggesting that higher discounts may reduce profitability.

5. Customer Analysis: Compared customer sales contribution and average order value. The Corporate segment recorded the highest average order value, at approximately 52,158.

6. Recommendations

1. Promote Electronics: Focus on Electronics promotions to increase sales and profit.

2. Control Discounts: Compare profit at different discount levels before offering large discounts.

3. Reduce Shipping Costs: Identify expensive shipments and negotiate better rates with delivery partners.

4. Improve Data Quality: Verify missing profit values and unusual financial records using original transaction data.